Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/17/2006
PI Name:	Atreya, Arvind Ph.D.		
Project Title:	Extinction of Gaseous Microgravity Diffusion Flames in Various Atmosphere		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous flames - premixed		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48109-2125	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	10/01/2003	**End Date:**	09/30/2006
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Im, Hong (University of Michigan)		
Grant/Contract No.:	NNC04AA10A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The overall task was divided into experimental and theoretical/computational components. The experimental effort consisted of using a porous spherical burner to issue various fuels (CH4, C2H4 & C2H2) and their mixtures (with CO2 & N2) into a controlled oxidizing environment (with various concentrations of O2, CO2, H2O and N2). These experiments provided data for understanding the flame chemistry at extinction via computations with detailed chemistry. The experiments are conducted in the 2.2s (and 5s) drop towers at the GRC. The supporting numerical calculations with detailed chemistry, soot and radiation are conducted at Michigan. The objective of this research is to experimentally and computationally investigate the conditions that will lead to extinction of gaseous diffusion flames in µg and understand the role played by gas and soot radiation in flame extinction. The scientific goal is to develop a fundamental understanding of the interaction between radiation and flame chemistry that is responsible for extinction of diffusion flames. The practical goal is to help develop fire suppression and detection technology for space applications. This		

	research is intended develop a fundamental understanding of the interaction between flame radiation and flame chemistry that is responsible for extinction of diffusion flames. The practical goal is to help develop fire suppression and detection technology for space applications. The results of work are also applicable to practical combustion systems where exhaust gas recirculation is used as a method of pollutant control in engines and furnaces.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will help develop fire suppression and detection technology for space applications. The results will also develop practical combustion systems where exhaust gas recirculation is used as a method of pollutant control in engines and furnaces. The results of work and the science developed are applicable to practical combustion systems where exhaust gas recirculation is used as a method of pollutant control in engines and furnaces.
Task Progress:	This project is divided into two parts: (i) An experimental component conducted at the NASA-Glenn drop tower facility, and (ii) A computational component conducted at the University of Michigan. Progress made in these components is summarized below: Experimental: Last year, final adjustments to measurement calibration and experimental control were made to the newly reconstructed experimental drop rig and safety approval was obtained. Several microgravity experiments were conducted in the 2.2 second drop tower facility using ethylene, a well documented sooty fuel, with total flow rate of 5cm3/s into air (21%O2, 79%N2) at 14.7psia. Studies of interest included investigating the role of fuel diluents CO2 and helium on radiative and chemical properties of the spherical diffusion flames. Fuel mixtures using ethylene (C2H4), and diluents CO2, He, and N2 were selected to maintain the temperature dependent energy storage term rho CP(T) approaches constant. Consideration was also made to balance CP while maintaining a constant fuel flow rate. This condition is important because the flame temperatures in both cases are nearly equal at the onset of the established flame. Temporal and spatial distributions of temperature, spectral radiation, and soot were measured with calibrated detectors. These experiments established the repeatability and accuracy of measurements and provided data for the effect of dilution on radiative and chemical flame properties. Considerable effort was spent on developing data processing techniques, programs and analysis tools. A new set of experiments are currently underway. In these experiments the atmosphere is modified by oxidizer side diluents. A new safety permit was obtained and safety protocols were satisfied. Currently experiments are being done with 18-50%O2 / 0-72%N2 / 0-40%CO2 / 0-52%He with ethylene as the fuel. Results of the fuel side dilution experiments enable us to understand of the effect of radiation absorption on micro-g flames. Visible flame images, flame growth rate, temperature field, spectral radiation measurements, and soot distribution measurements demonstrate the effect of CO2 as a diluent on the fuel side of the reaction zone. CO2 addition was found to decrease flame growth rate and reduce flame temperature for similar dilution despite efforts to balance the amount of fuel and temperature dependent properties, specifically CP, between test and control cases. Temperature data for enhanced CO2 on the fuel side indicates increased flame interior temperature which could be attributed to radiation reabsorption. Computational: First, the transient expansion of µg spherical diffusion flames was computationally simulated without including soot and flame radiation. Next, a one-dimensional spherical model was set up to study the transient ignition, combustion, and extinction process for highly radiating diffusion flames. Detailed chemical kinetic rates and transport properties were incorporated through the CHEMKIN and TRANSPORT packages and Newton iteration scheme via the TWOPNT module. A stiff differential-algebraic equations solver, DASPK, is also employed to find the transient flame dynamics. This model with detailed chemistry & radiation and the measurements will enable accomplishing the scientific goal of developing a fundamental understanding of the interaction between radiation and flame chemistry that will help fire suppression in space and pollutant-fee combustion on ground.
Bibliography Type:	Description: (Last Updated: 04/21/2017)
Abstracts for Journals and Proceedings	Chernovsky MK, Atreya A, Sacksteder KR. "Effect of Enhanced CO2 Concentration on Radiative Properties of Unsteady Spherical Diffusion Flames in Microgravity." 2005 Joint Meeting of the US Sections of the Combustion Institute, Philadelphia, PA, March 2005. 2005 Joint Meeting of the US Sections of the Combustion Institute, Philadelphia, PA, March 2005. , Mar-2005